

Tie-out roster: the whole credit suite

Every data point in both monitors, checked against the body that produces it. 5 September 2026.

857 of 862 data points tie. 5 differ. 0 could not.

Every figure on the ours side was read out of the shipped workbook — the cell a person opens — and never re-fetched. Every figure on the other side came off a document published by somebody else: a bank's own filed Call Report, or the agency that computes a series. The five that differ are set out first.

What that denominator is made of. Each bank has 69 raw fields, so the bank monitor holds 828 of them; 720 are compared here. The rest, per bank, are 8 ratios the FDIC computes from lines proved below and 1 field it publishes no quarterly figure for. The macro monitor holds 142 series and all 142 are compared, at their latest observation.

A correction to the first edition of this roster

The first edition said 776 of 778, and did not say what 778 was.

It was not every base data point. Each bank has 69 raw fields and only 53 were being compared; sixteen per bank went unmentioned. The firm asked directly whether every point had been done, and it had not.

The cause is why a tie-out can look complete while missing a fifth of its subject: **seven fields carried no citation at all**, and the tie-out only checks fields the provenance map cites. A check that examines what the map documents cannot discover what the map omits. Two further faults sat behind that one — bracketed expressions the reader silently returns nothing for, and income-statement codes cited without the prefix that finds them.

All eight fields are now compared, their citations are recorded, and a new guard checks that every citation parses and finds its line on a real filed Call Report. The eight FDIC-computed ratios per bank stay out of scope and are now named rather than omitted.

What does not tie

Five lines, all PNC Bank, and the workbook is not the side that is wrong.

Every quarterly net-charge-off field PNC reports differs from the bank's own filed report: credit cards by 515, C&I by 652, other consumer by 102, residential real estate by 6 and construction by 188 thousand dollars. The first edition found two of them, because the other three were not being checked at all. The workbook carries the FDIC's published quarterly figure to the dollar, and the FDIC's own year-to-date figure matches PNC's filing exactly. What does not reconcile is the FDIC's own arithmetic:

field	FDIC Q1	+	FDIC Q2	=	sum	FDIC year-to-date	gap
NTCRCDQ	43,842	+	47,617	=	91,459	91,974	515
NTCIQ	101,857	+	109,201	=	211,058	211,710	652
NTCONOTQ	15,470	+	10,994	=	26,464	26,566	102
NTRERESQ	1,113	+	1,880	=	2,993	2,999	6
NTRECONQ	-48	+	-571	=	-619	-431	188

and PNC's own filed Call Report agrees with the year-to-date column, not with the sum of the two quarters.

The same reconciliation across all twelve banks and all seven quarterly flow fields the FDIC publishes holds **79 times out of 84**. The five that fail are all PNC, and they are exactly the five lines above. No merger explains it: PNC's only 2026 acquisition events are branch transfers dated 6 July 2026, after the reporting date.

I believe the filing, because it is the document the bank signed and the FDIC's own annual figure agrees with it. Nothing was adjusted. The workbook faithfully carries what its source published, and this is written down rather than plugged.

What the tie-out found

Nine defects, and every one of them left the numbers correct, which is why a suite that now runs 541 tests had never seen any of them.

1. **A shipped workbook with a state missing from it.** Nebraska's house price index was blank. One `Internal Server Error` from the data provider blanked the slot and the build passed. The retry covered rate limits only. *Fixed.*
2. **Two series wearing each other's description** — the commercial real estate construction and nonfarm-nonresidential labels were swapped on the dashboard and in the raw tabs. *Fixed.*
3. **A tightening indicator with its alert switched off**, because it had been filed as a demand series, and demand series correctly get no alert. A real mortgage-tightening signal could never fire. *Fixed, and that turns an alert on.*
4. **Two more labels naming a different series** than the number beside them. *Fixed.*
5. **Four series declaring “billions” beside a figure in millions** — a factor of a thousand on the line a person reads. *Fixed.*

6. **The FDIC's own quarterly and annual figures for PNC do not reconcile**, on all five of its quarterly charge-off fields. Reported above; nothing changed, because our side is right.
7. **Seven fields had no citation at all** — the workbook landed a value and nothing recorded where it came from, so the tie-out never reached them. *Fixed: all seven cited and tied.*
8. **Bracketed expressions read as nothing**, silently. Every quarterly charge-off citation was affected, including one already flagged verified. *Fixed.*
9. **The filing parser discarded every ratio in every filing**, keeping whole numbers only. Dollar amounts are whole numbers, so nothing ever looked wrong; the capital ratios were in the data the whole time. *Fixed.*

And three defects in the checking itself, which is the likelier culprit than the code and was here too. My first pass cited the wrong Call Report line for C&I charge-offs (U.S. addressees only, where the measure is U.S. and non-U.S.), took the wrong column of the total capital ratio for the one bank that files two, and produced six blank source photographs that reported “ok”. Each showed up as an implausibly uniform failure across every entity — which is the tell.

The denominator, by set

Set	Checked against	Lines	Tie	Differ
JPMorgan Chase Bank NA cert 628	its own filed Call Report, 30 June 2026	61	60	1
Wells Fargo Bank NA cert 3511	its own filed Call Report, 30 June 2026	61	60	1
Bank of America NA cert 3510	its own filed Call Report, 30 June 2026	61	60	1
Citibank NA cert 7213	its own filed Call Report, 30 June 2026	61	60	1
US Bank NA cert 6548	its own filed Call Report, 30 June 2026	61	60	1
PNC Bank NA cert 6384	its own filed Call Report, 30 June 2026	61	55	6
Truist Bank cert 9846	its own filed Call Report, 30 June 2026	61	60	1
Capital One NA cert 4297	its own filed Call Report, 30 June 2026	61	60	1
KeyBank NA cert 17534	its own filed Call Report, 30 June 2026	61	60	1

Set	Checked against	Lines	Tie	Differ
Bank of New York Mellon cert 639	its own filed Call Report, 30 June 2026	61	60	1
Goldman Sachs Bank USA cert 33124	its own filed Call Report, 30 June 2026	61	60	1
Morgan Stanley Bank NA cert 32992	its own filed Call Report, 30 June 2026	61	60	1
FHFA All-Transactions house price indexes	Federal Housing Finance Agency	67	67	0
Charge-off and delinquency rates at commercial banks	Federal Reserve Board	29	29	0
G.19 consumer credit, the debt service ratios, FHFA monthly	Federal Reserve Board and FHFA	9	9	0
Senior Loan Officer Opinion Survey	Federal Reserve Board	13	13	0
S&P Cotality Case-Shiller house price indexes	S&P Dow Jones Indices	22	22	0
Z.1 commercial property price indexes	Federal Reserve Board	2	2	0
Total		862	857	5

Where the evidence is

One document per bank and one for the whole macro set. Each is self-contained: every screenshot is embedded, so it survives being forwarded to somebody who does not have this machine.

Document	What is in it
banks-12-2026-06-30/TIE-OUT-Bank-of-America-NA-cert3510-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Bank-of-New-York-Mellon-cert639-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it

Document	What is in it
banks-12-2026-06-30/TIE-OUT-Capital-One-NA-cert4297-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Citibank-NA-cert7213-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Goldman-Sachs-Bank-USA-cert33124-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-JPMorgan-Chase-Bank-NA-cert628-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-KeyBank-NA-cert17534-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Morgan-Stanley-Bank-NA-cert32992-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call

Document	What is in it
	Report that carries it
banks-12-2026-06-30/TIE-OUT-PNC-Bank-NA-cert6384-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Truist-Bank-cert9846-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-US-Bank-NA-cert6548-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
banks-12-2026-06-30/TIE-OUT-Wells-Fargo-Bank-NA-cert3511-2026-06-30.pdf	53 lines, each with the workbook cell photographed and the row of the filed Call Report that carries it
fred-142-series-2026-09-05/TIE-OUT-fred-142-series-2026-09-05.pdf	142 series across six publishers, with every workbook cell photographed and each agency's own page marked
keybank-card-30-89-2026-09-05/	the first exhibit: one figure traced link by link, and

What this does not prove

- **One quarter for the banks, one observation for the macro series.** The bank workbook holds sixteen quarters each and the macro workbook holds 13,905 observations. What was compared is the most recent of each — 778 figures, not the whole history. The Case-Shiller set was additionally checked a second month.
- **Raw inputs, not the ratios built on them.** Forty-eight of each bank's lines are dollar amounts read straight off the form, and two are ratios the bank files itself. The FDIC's computed ratios, and the dashboard's z-scores, bands and alert logic, are arithmetic over figures this proves — and are not themselves checked here. That was the instruction: the base data.
- **Twenty of the Case-Shiller series are tied through a ratio of two cells,** not a level, because S&P does not publish adjusted levels free. That pins the month-on-month move, not the absolute level.
- **The provenance map passed rather than was proved universal.** Every code it cites was found on every one of these twelve filings and every number matched. A citation that is right for these filers and wrong for a bank filing a different form would not show up here.
- **Nothing here was checked by a second person.**